

The number of parallelogram when are smaller parallegram is taken = 9
the number of parallelogram were two smaller parallelogram is taken = 12
The number of parallelogram when three smaller parallelogram is taken = 6
the number of parallelogram when four smaller parallelogram is taken = 4
The number of parallelogram when six smaller parallelogram is taken = 4
the number of parallelogram when nine smaller parallelogram is taken = 1
the number of total parallelogram

$\Rightarrow 9 + 12 + 6 + 4 + 4 + 1$

$\Rightarrow 36$

59. In a school every student is assigned a unique identification number. A student is a football player if and only if the identification number is divisible by 4, whereas a student is a cricketer if and only if the identification number is divisible by 6. If every number from 1 to 100 is assigned to a student, then how many of them play cricket as well as football?

(a) 4 (b) 8
(c) 10 (d) 12

Ans : (b) L.C.M. of (4, 6) = 12
then,
the number of student who play cricket as well as football $= \frac{100}{12} \Rightarrow 8.33$
≈ 8 (taken integer)

60. When a runner was crossing the 12 km mark she was informed that she had completed only 80% of the race. How many kilometres was the runner supposed to run in this event ?

(a) 14 (b) 15
(c) 16 (d) 16.5

Ans : (b) $\because$ 80% = 12 km

$\therefore 100\% = \frac{12}{80} \times 100 = \frac{12}{4} \times 5 = 15$ km

61. Raju has ₹ 9000 with him and he wants to buy a mobile handset; but he finds that he has only 75% of the amount required to buy the handset. Therefore, be borrows ₹ 2000 from a friend. Then

(a) Raju still does not have enough amount to buy the handset.
(b) Raju has exactly the same amount as required to buy the handset.
(c) Raju has enough amount to buy the handset and he will have ₹ 500 with him after buying the handset.
(d) Raju has enough amount to buy the handset and he will have ₹ 1000 with him after buying the handset

Ans : (a) $\because$ Raju has 9000 Rs
Required amount to buy the hand set

$\Rightarrow 9000 \times \frac{100}{75}$

$= 9000 \times \frac{4}{3}$

$= 3000 \times 4$

$= 12000$

$\because$ Raju borrowed 2000 Rs. from a friend.
$\therefore$ Raj has total amount (9000 + 2000)
= 11000
Hence,
Raju still does not have the required amount to buy mobile.

62. In 2002, Meenu's age was one-third of the age of Meera, whereas in 2010, Meenu's age was half the age of Meera. What is Meenu's year of birth ?

(a) 1992 (b) 1994
(c) 1996 (d) 1998

Ans : (b) In 2002,
Let age of Meera = 3x
$\therefore$ Age of Meenu = x
In 2010,
Age of Meera = (3x + 8)
Age of Meenu = (x + 8)
According to questions,
(3x + 8) = 2(x + 8)
3x + 8 = 2x + 16
x = 8
Hence, the age of Meenu in 2002 = x = 8 years
then Meenu's year of birth = 2002 – 8 $\Rightarrow$ 1994

63. Rakesh and Rajesh together bought 10 balls and 10 rackets. Rakesh spent ₹1300 and Rajesh spent ₹1500. If each racket costs three times a ball does then what is the price of a racket ?

(a) ₹ 70 (b) ₹ 90
(c) ₹ 210 (d) ₹240

Ans : (c) Let the price of ball = x
$\therefore$ the price of rocket= 3x
10 balls + 10 Racekts = 1300 + 1500

10 $\times$x + 10 + 3 x = 2800
40x =2800
x = 70Rs
then the price of rocket= 3x =3 × 70 = 210 Rs.

64. The ratio of a two-digit natural number to a number formed by reversing its digits is 4 : 7 The number of such pair is

(a) 5 (b) 4
(c) 3 (d) 2

Ans : (b) According to questions,
Requred numbers $\Rightarrow$ (12, 21) $\Rightarrow$ (24, 42)
$\Rightarrow$ (36, 63) $\Rightarrow$ (48, 84)
Hence,
the numbers of required pairs = 4

65. In an examination, A has scored 20 marks more than B. If B has scoured 5% less marks than A. how much has B scored?

(a) 360 (b) 380
(c) 400 (d) 420

Ans: (b) Let B obtained x marks,
then As obtained makrs= (x + 20)
According to question,

$5\% = \frac{20}{(x+20)} \times 100\%$

5 (x + 20) = 2000
5x +100 = 2000
5x = 1900
x = 380
Hence,
B obtained 380 marks

66. Seeta and Geeta go for a swim after a gap of every 2 days and every 3 days respectively. If on 1[st] January both of them went of a swim together, when will they go together next?

(a) 7[th] January (b) 8[th] January
(c) 12[th] January (d) 13[th] January

Ans: (d)
$\because$ Seeta go for a swim after a gap of every 2 days : that means Seeta go for a swim after 3 days. and Geeta go for a swim after a gap of every 3 days : that means Geeta go for a a swim after 4 days.
$\therefore$ The L.C.M. of (3, 4) = 12
$\because$ Seeta and Geeta went of a swim together 1st of January
$\therefore$ They go together next= (1 + 12)[th] January
=13th January

67. X, Y and Z are three contestants in a race of 1000 m. Assume that all run with different uniform speeds. X gives Y a start of 40m and X gives Z a start of 64m. If Y and Z were to compete in a race of 1000m. how many metres start will Y give to Z ?

(a) 20 (b) 25
(c) 30 (d) 35

Ans : (b) x = 1000m and x = 1000 m
y= (1000-40)m z = (1000-64)m
= 960 m z = 936 m
Hence, x:y:z= 1000 : 960:936
for y and z

$\frac{y}{z} = \frac{960}{936}$

$\frac{1000}{z} = \frac{960}{936}$

$z = \frac{936 \times 1000}{960}$

z = 975 m
Hence, y will give the start of 25 meters to z.

68. If x is grater then or equal to 25 and y is less than or equal to 40, than which one of the following is always correct ?

(a) x is greater than y
(b) (y–x) is greater than 15
(c) (y–x) is less than or equal to 15
(d) (x+y) is greater than or equal to 65

Ans : (c) $\because$ $x \geq 25$.....(i)
$40 \geq y$.....(ii)
From egn (i) & (ii),
$(x+40) \geq (y + 25)$
$(40–25) \geq (y–x)$
$15 \geq (y–x)$
Hence, (y – x) is less than or equal to 15

69. Ena was born 4 years after her parents' marriage. Her mother is three years younger than her father and 24 years older than Ena, who is 13 years old. At what age did Ena's father get married ?

(a) 22 years (b) 23 years
(c) 24 years (d) 25 years

Ans : (b)

	Father	Mother	Ena
When Era was 13 years old	37 + 3 = 40	13 + 24 =37	13
time of birth of Era	40–13 =27	37–13 =24	0
4 years before Ena's birth	27.4 $\Rightarrow$ 23	24 – 4 $\Rightarrow$ 20	Ena is not born yet

Hence, Era's father got married at the age of 23

70. Rakesh had money to buy 8 mobile handsets of a specific company. But the retailer offered very good discount on that particular handset. Rakesh could buy 10 mobile handsets with the amount he had. What was the discount the retailer offered ?

(a) 15 % (b) 20 %
(c) 25 % (d) 30 %

Ans : (b) $\because$ discount % $= \frac{\text{Free Items}}{\text{Total items}} \times 100\%$

$= \frac{2}{10} \times 100\% = 20\%$